

POC Idea: Intelligent QA Automation Assistant using Langflow (FundManager Application)

Overview:

A Langflow-based intelligent assistant designed to automate and streamline QA workflows for the FundManager application. The system will assist in generating test cases, validating workflows, performing accessibility checks, verifying notifications (email), and testing APIs through a unified conversational interface.

Objective:

To reduce manual QA effort by automating repetitive testing tasks and improving test coverage, accuracy, and efficiency—especially for fund-related workflows involving business rules, notifications, and integrations.

Current Challenges:

- Manual generation of test cases and uploading to Zephyr
 - Time-consuming accessibility testing processes
 - Manual verification of email notifications triggered by expense conditions
 - API testing requires separate tools and validation steps
 - Fragmented workflow across multiple QA tools
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Possibility to Automate:

- Auto-generate test cases based on user stories or requirements
 - Automatically upload test cases to Zephyr
 - Perform accessibility validation checks on UI flows
 - Validate email notifications triggered by fund/expense conditions
 - Execute and validate API responses
 - Provide a unified conversational interface for QA activities
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Approach:

- Build a tool-based agent using Langflow
- Integrate with:
 - Test management tools (e.g., Zephyr API)
 - Email systems for notification validation

- API endpoints for automated testing
 - Enable the assistant to:
 - Interpret QA scenarios and generate test cases
 - Execute predefined workflows (UI, API, notification checks)
 - Validate outputs and provide summarized results
 - Use modular components for:
 - Test case generation
 - Accessibility validation
 - Email parsing and verification
 - API testing
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Key Features:

- Conversational QA assistant
 - Automated test case generation and management
 - Accessibility testing support
 - Email notification validation
 - API testing and response validation
 - Integration with QA tools ecosystem
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Sample Use Cases:

- Generate test cases for fund expense validation scenarios
 - Upload generated test cases directly to Zephyr
 - Validate if email notifications are triggered for threshold breaches
 - Check accessibility compliance for fund management UI
 - Test APIs related to fund transactions and verify responses
 - Query test execution results via chat
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Expected Outcome:

- Reduced manual QA effort
- Faster test case creation and execution
- Improved test coverage (UI, API, notifications, accessibility)

- Centralized QA workflow through a single interface
 - Foundation for scaling into a fully autonomous QA assistant
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Future Enhancements:

- Add RAG-based knowledge retrieval for requirements and past defects
 - Integrate with CI/CD pipelines for automated test execution
 - Support for regression suite automation
 - Advanced analytics and reporting dashboards
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